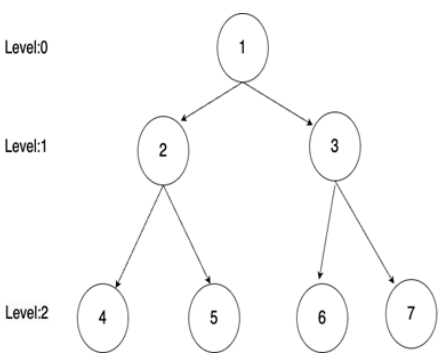


Write a recursive method `leftRightAddition()` that takes the root of a binary tree as a parameter. The method will add the summation of the left subtree of the given root to the summation of the right subtree of the given root. But while calculating the summation of a tree, for even level nodes, the value will be taken as a positive value, and for odd level nodes, a negative value will be taken. Consider the Node class for the binary tree already defined with `data`, `left`, and `right` fields. You may use helper methods. **YOU CAN NOT USE ARRAYLIST OR ARRAYS. YOU CAN NOT USE ANY BUILT-IN METHODS**

Sample Input:	Sample Output	Explanation
	17	Summation of left subtree + Summation of right subtree $= ((-1)*2+4+5)+((-1)*3+6+7)$ $= -2+4+5-3+6+7$ $= 17$

Section B: MCQ and Short Questions

[6 marks]

MCQ: Circle the correct answer.

[1 mark each]

1. Which traversal of a binary tree visits the root node between visiting the left and right subtrees?

- a) Pre-order b) In-order c) Post-order d) Level-order

2. What is the maximum number of nodes at level k in a binary tree (where the root is at level 0)?

- a) k b) $2k$ c) 2^k d) k^2

3. For a full binary tree with height h (where a single-node tree has height 0), which of the following correctly states the number of leaf nodes?

- a) h b) $h+1$ c) 2^h d) $2^{(h+1)} - 1$

4. Consider a binary tree where post-order traversal gives: 42, 56, 23, 61, 77, 34, 14. Which node is the root?

- a) 42 b) 23 c) 34 d) 14

5. Which of the following properties is true for every node in a binary tree (not necessarily a BST)?

- a) Each node has exactly two children
b) Each node has at most two children
c) Left child value is always smaller than right child value
d) All leaf nodes are at the same level

Short Question: [2 marks]

6. In a recursive function that traverses a binary tree, what is the base case that stops the recursion?

Answer: